Cournot-Cold: A Calibrated Forecaster You Can Check

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Three models, seventeen days, $62 of GPU, and every result that died

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Abstract

Cournot-Cold is a family of three conditional probability estimators, at 8B, 4B and 1.7B parameters, which forecast binary outcomes from a question, its resolution criteria and an as_of timestamp while conditioning on no retrieved evidence. On a contamination-free split of 277 questions that resolved after a fixed freeze, the 8B scores a Brier of 0.1893 against 0.2500 for a constant at the base rate. The models are not state of the art and this report does not argue that they are; they cost $62 of GPU across seventeen days. The contribution is measurement. We separate two sources of variance that the literature routinely conflates, and find that two runs of an identical recipe at an identical seed differ by +0.0051 [+0.0019, +0.0082] while two runs at different seeds do not differ at all. That sets a threshold of approximately ± 0.008 for any comparison between separately trained models, and it invalidated the intervals attached to most of our own model-versus-model claims. Across a 4.7-fold parameter range calibration is flat while resolution falls, so scale buys discrimination rather than trustworthiness. We report calibration bin-free, publish 6,734 forecasts frozen before any of their outcomes existed, and give a failure record of fifty-one dated corrections grouped by the six mechanisms that produced them, of which four are now prevented by construction and two are not. Every number here is recomputable offline from the shipped forecasts, with no model weights, no GPU and no network.

Changelog and errata

Dated, additive, never subtractive. Struck text stays struck and nothing is deleted. This section sits at the front rather than the back because a correction a reader has to go looking for is a correction that does not do its job.

date	entry
2026-08-29	**v1.0.** First publication.
2026-08-29	~~"Cournot-Cold 8B Wide is the next release."~~ **Declined.** Its Brier advantage on Polymarket decomposes into calibration rather than resolution, and a post-hoc map buys the same thing. See §6.1, arm mixed_full.
2026-08-28	~~"Cournot-Cold 1.7B dev Brier is 0.1752."~~ **It is 0.1803.** The shipped model is the second of two runs of an identical recipe, and the two differ significantly. See §3.2.
2026-08-28	~~"±0.003 is this project's noise floor for comparing two models."~~ **±0.003 is the standard deviation of a single run.** The threshold for a difference between two separately trained models is **approximately ±0.008**. This reaches back over most of the model-versus-model claims the project has made. See §3.2.
2026-08-27	~~"El Lahib et al. found 17.09% of news queries leaked the outcome, contaminating 57% of the benchmark."~~ **Fabricated.** The attribution, both figures, and the identity of the paper were all wrong. Corrected figures appear in §6.2, mechanism E.
2026-08-27	~~"The published split's date leak inflated resolution and cost calibration."~~ **Not supported.** The paired test is null and underpowered by a factor of 2.2 to 3.5. The fix stands on the non-negotiable rule it violated, not on a measured effect. See §6.2, mechanism C.
2026-08-26	~~"The model matches the crowd at the market's half-life."~~ **Crossover is φ* = 0.390 [0.27, 0.51].** At $\varphi = 0.50$ the model was already behind.

Every entry above has a dated decision record behind it in the research repository's log, written at the time and in the order it happened. That ordering is the only thing that makes a changelog evidence rather than presentation.

0 What this is

Cournot-Cold is a family of three conditional probability estimators, at 8B, 4B and 1.7B parameters. Each takes a question, its resolution criteria, a resolution date and an as_of timestamp, and returns a probability in [0,1]. They condition on no evidence at all. That is the tier's definition rather than a limitation of the implementation: the family has three tiers, and Cold is the one that answers from the question alone.

The models are not state of the art, and this report does not argue that they are. The nearest published peer, OpenForecaster-8B [1], uses the same base model with a retrieval loop and roughly 1,000 H100-hours behind it. This family cost **$62 of**

GPU across seventeen days, and where the two are comparable it is not obviously ahead.

The contribution is the measurement apparatus and the record of what it caught. Four things in particular:

1. **Two variances, kept apart.** Almost every model comparison in the literature reports sampling variance over the evaluation set, or reports nothing at all. We measured the other one, by retraining the same recipe, and it is larger. It invalidated the interval attached to most of our own model-versus-model claims. See §3.2.
2. **A pre-registered failure record with a denominator.** Fifty-one corrections and retractions, grouped by the mechanism that produced them, each carrying the guard that caught it or an admission that none existed. See §6.
3. **A forward track record.** 6,734 forecasts frozen and published before any of their outcomes existed. See §8.
4. **Every number in this report recomputable offline from what ships**, with no GPU, no model weights and no network. See §10.

The honest summary of the modelling result is one sentence. A scalar Brier head on a Qwen3 base, trained on 81,870 terminal outcomes with no reasoning traces and no teacher, is well calibrated across a 4.7-fold parameter range, and what falls with scale is resolution rather than calibration.

The honest summary of everything else is a second sentence. Most of what we believed at any given moment in those seventeen days was wrong in a way that a gate later caught, and the gates that caught the most were the ones that had never fired before.

1 Scope: what this report does not claim, and what it does not disclose

This section sits second, before any number, following the convention that GPT-4's technical report established for the slot. That report uses the slot to say what it will not disclose. We use it for both purposes, because for a report whose subject is measurement the more useful half is the first.

Not claimed

- **Not that these models beat any commercial forecaster, prediction market, or human aggregate.** On the axis where we can check it the market wins. Its Brier runs from 0.1854 to 0.0974 across a question's life, and the model crosses below it only at $\varphi^* = 0.390$ [0.27, 0.51].
- **Not that the 4B wins against the 8B.** It is indistinguishable from it within a

stated margin, which is a different and weaker claim. §3.5 states the margin and justifies it.

- **Not that calibration transfers across venues.** It does not. ECE is 0.023 on the training venue's dev split against 0.156 on Kalshi.
- **Not that any of this generalises** past binary judgmental questions resolved by a named venue, at a single as_of, from a single model with no ensembling.
- **Not that the corrections in §6 are the complete set.** They are the ones that were caught. The distribution in §6.3 is a lower bound whose direction is knowable.

Not disclosed, and why

- **The raw corpus is not redistributed.** Question text belongs to the venues. Every evaluation row ships keyed by the venue's own stable question id, so every number remains reproducible without us redistributing anyone's content.
- **Weights are on the Hugging Face Hub** rather than in the artifact repositories.
- **The decisions log is not published in full.** It contains vendor pricing and unreleased directions. §6 and Appendix A publish its structure, its counts, and every entry that bears on a number in this report.

One convention, used throughout

Any figure this report has not verified against its own artifact is marked **UNVERIFIED**. That mark appears six times below. It is not a hedge. It is the difference between a number we checked and a number we are repeating.

2 The artifact and the interface contract

Every model in the family satisfies one contract:

```
input:  question + resolution criteria + resolution date + as_of  [+ evidence, for Live/Deep]
output: p  in  [0,1]
```

Cold takes no evidence. as_of is required and has no default. A forecast without an information boundary is not a forecast, and a default is how such a boundary silently becomes "whenever someone happened to run the job."

	Cournot-Cold 8B	**Cournot-Cold 4B**	**Cournot-Cold 1.7B**
Base	Qwen3-8B	Qwen3-4B	Qwen3-1.7B
Base release	2025-04-29	2025-04-29	2025-04-29
Adapter	LoRA r=32, α=64	same	same

	Cournot-Cold 8B	Cournot-Cold 4B	Cournot-Cold 1.7B
Adapter size	349 MB	264 MB	160 MB
Trainable params			34,867,200 (1.99%)
Training corpus	81,870 terminal outcomes	same	same
Seed	20260822	same	same
Train wall-clock	3h31m	2h34m	1h32m
Head	Qwen3ForSequenceClassification, num_labels=1	same	same
Objective	Brier loss on $\mathrm{sigmoid}(\mathrm{logit})$	same	same

Openness class. Model Openness Framework [2] Class II, Open Tooling. Weights, inference code, evaluation code, evaluation data, model card and datasheet are released. Training data is not redistributed because it is venue content, and training code ships but the corpus build depends on a licensed dump. Stating the class is a falsifiable claim; the adjective "open" is not.

Licensing. Code under Apache-2.0. Weights, forecasts and evaluation metadata under CC BY-NC 4.0. Question text is not redistributed. This was wrong at first release: the code shipped under no grant at all, which made it published and legally unusable. It was corrected on 2026-08-29.

A note on the head, because it is the whole design

A causal model wrapped for sequence classification pools the last non-pad token, so the prompt's final hidden state becomes a single logit. Three consequences follow, and together they are why this shape was chosen over text supervised fine-tuning.

The objective is the metric. Brier loss on a probability output is a proper scoring rule evaluated directly. Next-token cross-entropy over the digit string 0.31 is not, because 0.31 and 0.29 share no gradient relationship.

There is no decoder to collapse. The first text-SFT arm returned 392 of 490 forecasts as exactly 0.0. A scalar head has no argmax.

The reasoning is never in the model's input, so the corpus needs no distilled traces, no teacher and no generation pass. That is what let training reach the whole 81,870-question split at zero marginal data cost, where the trace pipeline had capped earlier arms at 7,381 examples.

One implementation detail is worth stating because getting it wrong is silent. Sequence classification reads the last non-pad token, so the model trains with **right padding**. Left padding would place the pad run after the content and pool the wrong position. This is the opposite of the generation case. Under the mistake the model still trains, the loss still falls, and every forecast is read off a pad token.

3 Measurement before modelling

This section comes before the results and is longer than them. That ordering is the report's argument. A number's construction is load-bearing, and putting it after the number invites the reader to accept the number first.

3.1 What the evaluation is, and what it refuses

Three splits, keyed on actual resolution time (resolved_at), never on the open date and never on the scheduled resolution date. A question scheduled for March that resolved in June resolved in June.

```
FREEZE     = 2026-08-15T00:00:00Z
DEV_START  = FREEZE - 365 days

TRAIN      resolved_at <  DEV_START           81,870 questions
DEV        DEV_START   <= resolved_at <= FREEZE   n = 3,000 evaluation subset
PUBLISHED  FREEZE      <  resolved_at          n = 277
```

dev and published are different artifacts with different status, and that distinction is the one thing in the harness enforced in code rather than in prose. dev is carved from the training side and may be contaminated through base pretraining, so it gates iteration and is never published. published is contamination-free by construction, because every question in it resolved after a freeze that precedes it, and it is the only source of an external number.

The guard is a refusal rather than a warning:

```
if not split.publishable:
    raise NotPublishableError(
        f"refusing to build a card-bound artifact from split {split.value!r}: "
        "only 'published' is contamination-free by construction."
    )
```

Three properties of that guard are deliberate, and each was argued before it was written.

There is no override parameter. No force=True. The whole point of the split is that this path cannot be taken on dev data, and an override would make the guard a formality that someone under deadline would eventually find.

The split is derived from the records rather than read from a field. The evaluation slice holds records and re-derives the split from resolved_at at emission time. There is nothing to set wrongly and nothing to remember.

A record landing exactly at the freeze goes to dev. Losing one question costs nothing. Weakening the claim that contamination is impossible costs the product.

Two further refusals sit alongside it. A slice spanning both splits is refused rather than partitioned silently. And a run that has discarded its as_of values is refused, because forecast horizons are resolved_at − as_of and nothing can reconstruct them afterwards. That is an omission which fails at no point where anyone is looking.

Certification. A model card number must come through certify, which is the only route to a certified card. It makes three refusals. The first is a manifest split that differs from the card split, which catches a manifest built for a different run and pasted onto this one. The realistic way provenance goes wrong is not fabrication but reuse. The second is a manifest freeze that differs from the card freeze. The third is a base model released after the freeze, and that one matters most: such a model may have trained on the very questions the published split is built from, and nothing else in the pipeline would notice, because the leakage detector screens documents and this failure involves no document. We key on the base model's release date rather than its stated cutoff. A cutoff is self-reported, frequently understated, and unverifiable from outside. A release date is a fact.

Question flow, following the CONSORT diagram that clinical reporting made standard and machine learning has not adopted:

```
Manifold binary markets in snapshot          187,113
  |- rejected at normalization                    (counted, categorized)
  \- normalized, well-formed                 ~99,000
        |- TRAIN  (resolved_at < DEV_START)       81,870  -> training
        |- DEV    (DEV_START <= resolved_at <= FREEZE)
        |       \- stratified evaluation subset    3,000  -> gating only
        \- PUBLISHED (resolved_at > FREEZE)
                |- unresolved or other                28  -> dropped, counted
                |- no close_ts                         0  -> would be refused
                \- eligible                          277  -> every external number
```

That final figure, n = 277, is the denominator behind every headline number in this report.

3.2 The two variances, kept apart

This is the section the report exists for.

Two distinct sources of variation can move a Brier score. The literature routinely reports one, sometimes neither, and almost never both with labels [3, 4, 5], and the consequence of ignoring the second is well documented in reinforcement learning [6].

	what it is	how it is estimated	what it licenses
Question-sampling variance	would a different draw of questions have given a different score?	paired, question-clustered bootstrap; 10,000 resamples; seed 20260822; percentile method	comparing two **fixed forecast sets**
Training-run variance	would re-running the same recipe have given a different model?	re-run the recipe unchanged and score it	comparing two **separately trained models**

We spent most of the project reporting the first and quoting it against claims that needed the second.

How we found out. The 1.7B's weights were destroyed before a ship-or-no-ship decision had been made, which is its own failure and is recorded in §6.2 under mechanism F. Regenerating them, we gated the retrain on reproducing the destroyed run's forecasts. Same corpus, same seed 20260822, same learning rate, same schedule, same footing, same base model. The tolerance was 0.02, chosen before any data existed.

The gate fired.

run	dev Brier (n = 3,000)
1	0.1752
2 (shipped)	0.1803
3	0.1805

mean 0.1787 · sd 0.0030 · range 0.0053

Run 2 minus run 1, paired and question-clustered, is **+0.0051 [+0.0019, +0.0082]**, which is significant. Per-question forecast differences reached a maximum of **0.6793** against a tolerance of 0.02.

That is larger than our measured cross-seed difference. Both shipped models replicate across different seeds at +0.0012 [−0.0016, +0.0041] and +0.0021 [−0.0009, +0.0052], and both are null. Two runs at the same seed differ significantly while two runs at different seeds do not. The seed pins the data order and the initialisation. It does not pin the result.

The correction this forces. The project had quoted ±0.003 as its noise floor. That figure is right, and it was being used for the wrong quantity.

- **±0.003 is the standard deviation of a single run.**
- The difference between two independently trained runs has sd $\approx 0.003 \times \sqrt{2} \approx$ **0.0042**.
- So the 95% threshold for a real difference between two separately trained models is approximately ±**0.008**, not ±0.003.

A Brier difference between two separately trained models is not a finding below approximately 0.008, whatever interval is printed beside it.

Figure 1 plots the three runs against the ±0.008 band, with the cross-seed replications shown alongside for contrast. The point of the figure is the comparison between the two: the same-seed spread is wider than the cross-seed spread.

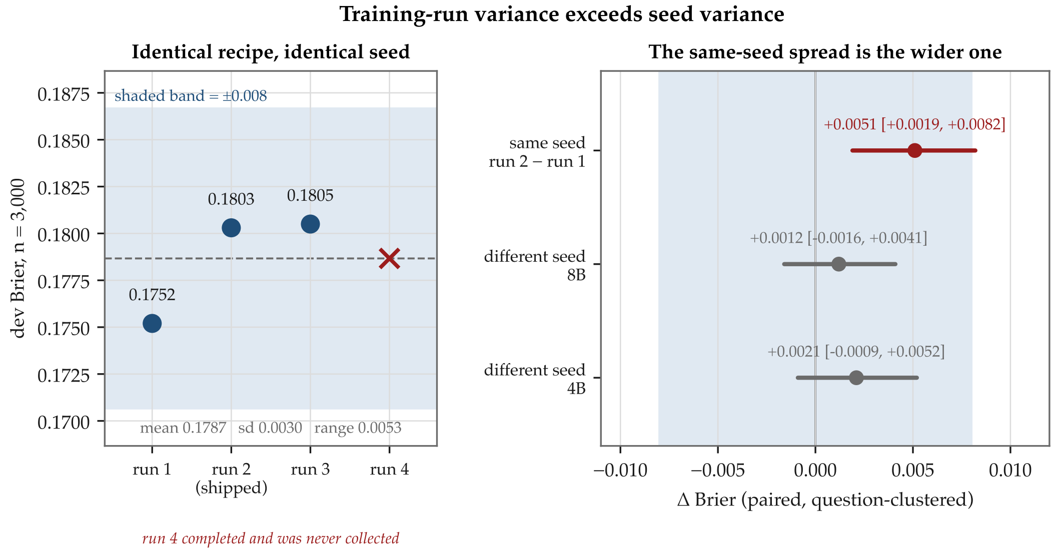

Figure 1: **Training-run variance exceeds seed variance.** Left: three runs of an identical recipe at an identical seed, against the 0.008 equivalence margin. Right: the same-seed difference is significant while both cross-seed differences are null.

What this reaches, and what it does not. The paired bootstrap remains correct for everything that is not a comparison of two training runs: model against market, model against a constant, one model across two splits, calibration against resolution within a single set of forecasts. Nulls are unaffected and in fact strengthened, since a wider true uncertainty makes "indistinguishable" easier to support rather than harder. What the correction reaches are positive claims, and it retired one on a public model card that had described a comparison as well-powered on exactly the mistaken reading above.

Four limits, stated and kept intact.

1. **n = 3.** A standard deviation on three points is itself very noisy. The range, 0.0053, is the more robust statistic and is quoted alongside.
2. **A fourth run completed and was never collected.** A pod was terminated from a web console thirteen hours after the watcher fired. See §6.2, mechanism F. The measurement stands at n = 3 because of an operational failure rather than a design choice, and redoing it costs roughly $0.70 and an hour and a half.
3. **Runs 1 and 2 were on different machines while runs 3 and 4 shared one.** Holding hardware fixed can only reduce apparent variance, so the pooled spread

is a lower bound on what a fresh run would show. That is the conservative direction.

4. **The mechanism is unverified.** Non-deterministic CUDA kernels is the hypothesis, since scalar_train.py does not call torch.use_deterministic_algorithms, but it is written here as a hypothesis. The CUDA version is ruled out: runs 2 and 3 agree across different versions while run 1 differs from both despite sharing one with run 3. **UNVERIFIED.** Testable for roughly $0.70, and not run.

Determinism is deliberately not forced. The quantity of interest is how different the model is when the recipe runs again. Forcing deterministic kernels would measure something else, namely the reproducibility of one artifact rather than the variance of a recipe. Both matter, and this measurement is only about the second.

Variance components still unmeasured, listed so the set above is not mistaken for complete: prompt-template sensitivity, serving-stack version, and the interaction between corpus ordering and the OneCycle schedule. Our intervals lower bound the actual variation.

3.3 Interval construction, stated

- **Method.** Percentile bootstrap over cluster indices. Chosen over the normal approximation because Brier differences are skewed at this sample size, and over BCa because its acceleration term needs a jackknife over clusters that costs more than the interval is worth here.
- **Resamples.** 10,000, enough that the endpoints are stable to about the third decimal.
- **Seed.** 20260822, fixed. A bootstrap that moved between runs would make two reports of the same result disagree.
- **Clustering.** By question_id. Where a run scores one question at five as_of placements, that is 629 forecasts but only 127 independent things. Resampling forecasts independently would shrink the interval by roughly $\sqrt{5}$.
- **Pairing.** Differences are taken first and bootstrapped second. Resampling the two arms separately would put question difficulty, which is much larger than any effect here, back into the estimate. Arms scored on different question sets are intersected on question_id before anything else happens.
- **What the bar captures.** Question-sampling variance only. It does not capture training-run variance (§3.2), prompt sensitivity, or serving-stack version.

Resolution needs its own machinery, and this is easy to get wrong. Brier is a mean of per-observation losses, so a paired difference has a per-observation form. Resolution does not. It is a binned aggregate over the whole arm with no per-observation value to difference, so it gets its own resample loop that redraws questions with replacement and recomputes the decomposition on each draw, at the same seed and resample count. Substituting the Brier interval for it is a category error we have made and caught.

An open decision, recorded rather than resolved. Below a few hundred questions the bootstrap itself undercovers, and Wilson or Bayesian intervals are indicated [7]. Our published split is n = 277. The intervals on published-split comparisons in this report should therefore be read as slightly optimistic in coverage. **UNVERIFIED**, in that we have not measured the undercoverage at our own n.

3.4 Power, and the minimum detectable effect

A null whose confidence half-width is wider than the quantity being measured is no evidence, not evidence of no effect. Every null in this report carries its half-width and, where it matters, the n that would be needed.

comparison	split	n	effect	half-width	verdict
4B − 8B	dev	3,000	+0.0010	0.0030	equivalent within margin (§3.5)
4B − 8B	published	277	−0.0062	0.0111	**underpowered**; half-width is 1.8× the effect, and n ≈ 888 would be needed
1.7B − 8B	dev	3,000	+0.0078	0.0038	significant, but see §3.5
1.7B − 8B	published	277	+0.0111	0.0135	**underpowered**

At n = 500 the resolution half-width was approximately 0.012, against resolutions of 0.008 to 0.015, which made it literally unmeasurable. At n = 3,000 it fell to roughly 0.004. That calculation is why the dev evaluation set holds 3,000 questions rather than 500, and it was performed after two comparisons had been reported as findings and died within the hour.

The rule is symmetric, and it is stated that way so it cannot be applied selectively. A difference below the threshold is unresolved, not disproven. It does not retire an inconvenient result any more than it establishes a convenient one.

3.5 The equivalence margin

The headline comparison in this family is that the 4B is indistinguishable from the 8B. That is an equivalence claim, and it had been reported the way equivalence claims are usually reported in machine learning, which is as a null. Medicine has a reporting standard for exactly this case, because the failure mode is notorious enough to have its own citation.

Absence of evidence is not evidence of absence [8]. The CONSORT extension for equivalence and non-inferiority trials [9] requires three things a null does not. The margin must be defined, it must be justified, and the conclusion must be stated as an interval relative to the margin. An audit of published trials [10] found the margin defined in 94% of them and justified in about a quarter.

We can produce the justified version, because we measured the floor.

> ***Equivalence margin:*** ***δ = 0.008 Brier.*** *Derived as the 95% threshold for a difference between two separately trained models, which is itself derived from a measured three-run replication of the identical recipe (sd 0.0030, range 0.0053; see §3.2).* ***4B − 8B on dev, n = 3,000: +0.0010 [−0.0020, +0.0040].*** *The interval lies entirely within $\pm\delta$. The claim is equivalence within the run-to-run noise of the training procedure. It is not "no difference", and it is not "the 4B is better".* ***4B − 8B on published, n = 277: −0.0062 [−0.0175, +0.0048].*** *The interval is wider than δ on both sides. This is not an equivalence result. It is underpowered, and the two must not be reported as though they were the same finding.*

Separating the axes that meet the margin from those that are merely underpowered is precisely what gets lost when an equivalence claim is dressed as a null, and a reader who knows the literature will check for it.

The 1.7B, judged the same way. 1.7B − 8B on dev is +0.0078 [+0.0040, +0.0116]. The point estimate sits just inside $\delta = 0.008$ and the interval crosses it. Stated honestly: the 1.7B is worse than the 8B by an amount comparable to the noise of retraining either of them. It ships anyway, and §5 says why.

4 Calibration reporting

The product is a calibrated probability, so this is where the report is most technical.

4.1 The bin-free decomposition, as primary

Binned expected calibration error has a problem the field mostly declines to state. The binning scheme moves the number materially, and neither choice is right. Equal-width binning leaves bins too sparse to estimate a frequency on a skewed set. Equal-mass binning has edges that move between checkpoints, so two runs are binned differently and then compared anyway. Our cards previously shipped two ECE numbers because we could not defend one, which discloses a problem rather than solving it.

The CORP decomposition [12] removes the choice. The reliability curve is the isotonic fit obtained by pool-adjacent-violators, so there are no bins, no scheme to defend, and no edges that drift when the data changes.

$$\text{score} = \text{MCB - DSC + UNC}$$

MCB, miscalibration, is how much the score improves under the optimal monotone recalibration. Zero means already calibrated. It is always non-negative, because recalibration cannot hurt on the data it was fit to, which is also why it is an in-sample quantity rather than a held-out claim.

DSC, discrimination, is how much the recalibrated forecast beats the constant. It is the bin-free counterpart of Murphy resolution [11].

UNC, uncertainty, is the score of a constant forecast at the base rate. It is a property of the evaluation set rather than of any model, and it must be reported alongside the other two or cross-venue comparisons become uninterpretable.

The residual is exposed as a computed property rather than a stored field, so a caller can assert it rather than trust it. It is $0.0\mathrm{e}{+}00$ on every split of every shipped model.

The shipped ladder on the published split, n = 277, base rate 0.4946, UNC 0.2500:

model	**Brier**	**MCB ↓**	**DSC ↑**	**dev Brier (n = 3,000)**	**dev cal**	**dev res**
Cournot-Cold 8B	0.1893	0.0095	0.0702	0.1674	0.0010	0.0718
Cournot-Cold 4B	**0.1831**	0.0124	**0.0793**	0.1684	0.0007	0.0708
Cournot-Cold 1.7B	0.2030	0.0106	0.0576	0.1803	0.0004	0.0593

The ladder's actual finding is in the last two columns rather than the first. Calibration is flat across a 4.7-fold parameter range, at 0.0010, 0.0007 and 0.0004, all inside the noise floor. Resolution falls monotonically, at 0.0718, 0.0708 and 0.0593. Scale buys discrimination, not calibration. For a product whose value proposition is a trustworthy probability rather than a sharp one, that is the more useful of the two things to learn, and it argues directly for shipping the small model.

Figure 2 renders MCB, DSC and UNC as stacked bars per model, because that makes improving Brier by hedging toward the base rate visually impossible to hide. That failure mode is invisible in an aggregate Brier, and it is the one this family is most exposed to.

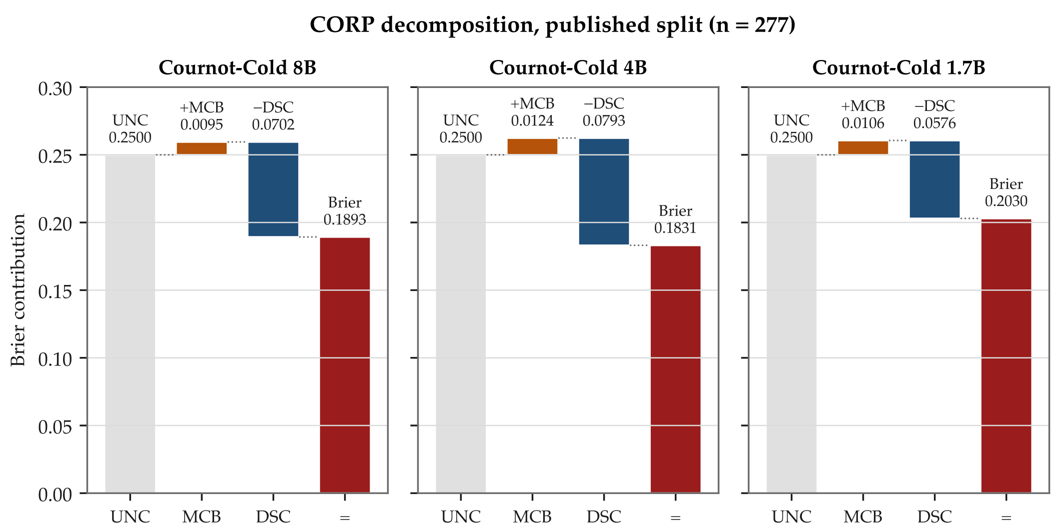

Figure 2: **Calibration is flat across the ladder; discrimination is not.** The CORP identity drawn as a waterfall: start at the uncertainty of the evaluation set, add miscalibration, subtract discrimination, land on the Brier.

4.2 Binned ECE, under both schemes, with its bound stated

Binned ECE is reported anyway, because readers expect it, and it is reported under both schemes with the spread visible. It is also framed correctly: binned ECE is a lower bound on the true calibration error, and it rises with bin count [13]. A single ECE number at an unstated bin count is not a measurement.

For the 8B on the published split, equal-width at 10 bins gives 0.0558 and equal-mass at 10 bins gives 0.0525. The spread between schemes, 0.0033, is itself of the same order as several effects this project has argued about, which is the case for CORP in a single number.

The harness raises a CoarseBinningWarning when the decomposition's residual is a material fraction of the score, because the Murphy identity closes regardless and that condition is otherwise invisible.

4.3 The output histogram, as a first-class metric

Collapse toward 0.5, or onto a handful of round values, is the characteristic failure mode of a trained forecaster, and it looks fine on every average metric in the suite. So the histogram is reported rather than inspected. **Figure 3** shows it for the 8B on the published split.

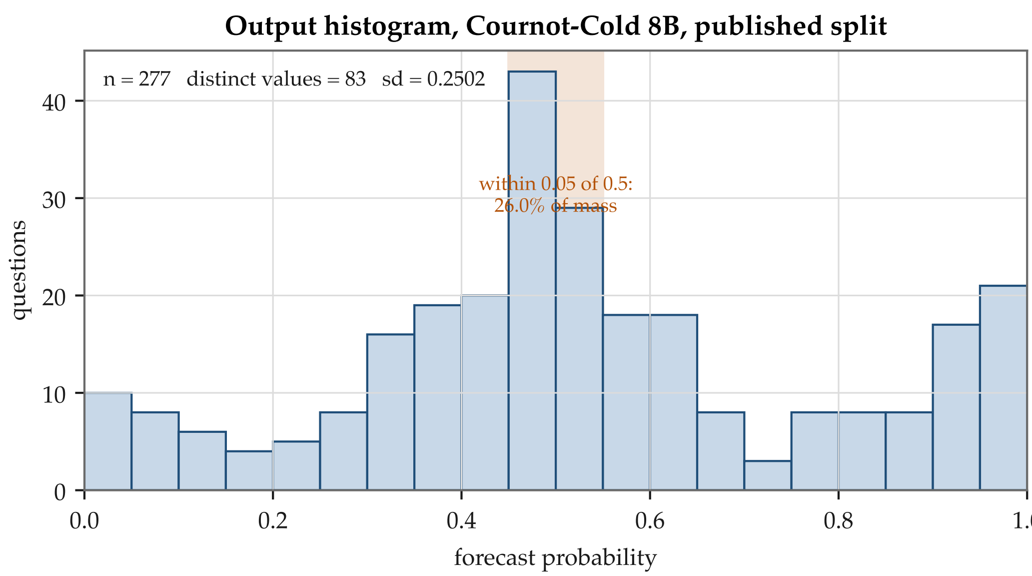

Figure 3: **The output histogram is a first-class metric, not a diagnostic.** Collapse toward 0.5 is the characteristic failure of a trained forecaster and is invisible in every average in the suite.

The 8B produces 83 distinct values with sd 0.2502 on the published split. The 1.7B produces 1,251 distinct values of 3,000 on dev, with sd 0.2739, 6.34 bits of entropy, and 15.7% of mass within 0.05 of 0.5.

This gate has fired for real, three times. The first text-SFT arm produced 392 of 490 forecasts at exactly 0.0. A later arm produced 2,727 of 3,000, or 90.9%, at 0 or 1, while its aggregate Brier looked merely bad rather than degenerate. And the varied-$\mathrm{as\_of}$ arm, the one that failed its pre-registered gate, narrowed from sd 0.2868 to 0.2597 with mass near 0.5 rising from 14.4% to 17.0%. That is what hedging looks like when it is dressed as robustness.

4.4 Reliability by forecast horizon

Two horizons exist in this project, and they are never interchangeable. $\mathrm{question\_lifetime}$, which is $\mathrm{resolved\_at} - \mathrm{open\_date}$, is a property of the record and is what the corpus census buckets on. $\mathrm{forecast\_horizon}$, which is $\mathrm{resolved\_at} - \mathrm{as\_of}$, is a property of the forecast and is what reliability diagrams break out by.

Reliability diagrams break out by the second. **Figure 4** shows the shipped 8B on the published split. Two rendering choices in it are deliberate. Bin counts appear as marker area with a count strip beneath, because a bin holding three questions plots identically to one holding three thousand and the eye reads a wobble in a three-question bin as miscalibration. And empty bins stay empty, so the polyline breaks across them rather than interpolating a segment nobody measured.

4.5 Post-hoc calibration, measured and then declined

Isotonic, beta and temperature maps are all implemented, and all are cross-fitted by question rather than by supervision point. Points from one question share an outcome, so splitting by point would put the same outcome on both sides of the fold and reintroduce exactly what cross-fitting removes.

The result is a negative one. On the shipped models, post-hoc calibration does not help. For the 8B on dev it makes things marginally worse, moving 0.1674 to 0.1681. Temperature scaling, which is exactly resolution-preserving [14], was measured at ten times below seed noise. The map is retained in the harness and not applied to the shipped models, and the reason is stated rather than left as an omission: a scalar Brier head trained directly on outcomes is already doing what the map would do.

One caveat limits every cross-fitted calibration number here. The folds are drawn from one twelve-month window and one question distribution, so the result is an upper bound on what calibration buys rather than an estimate of it.

5 What worked, in order of measured effect

Deliberately shorter than §6. Every row carries an interval, the read it belongs to, and the variance class it was judged against. **Figure 5** plots the rows that have intervals against the ± 0.008 training-variance band.

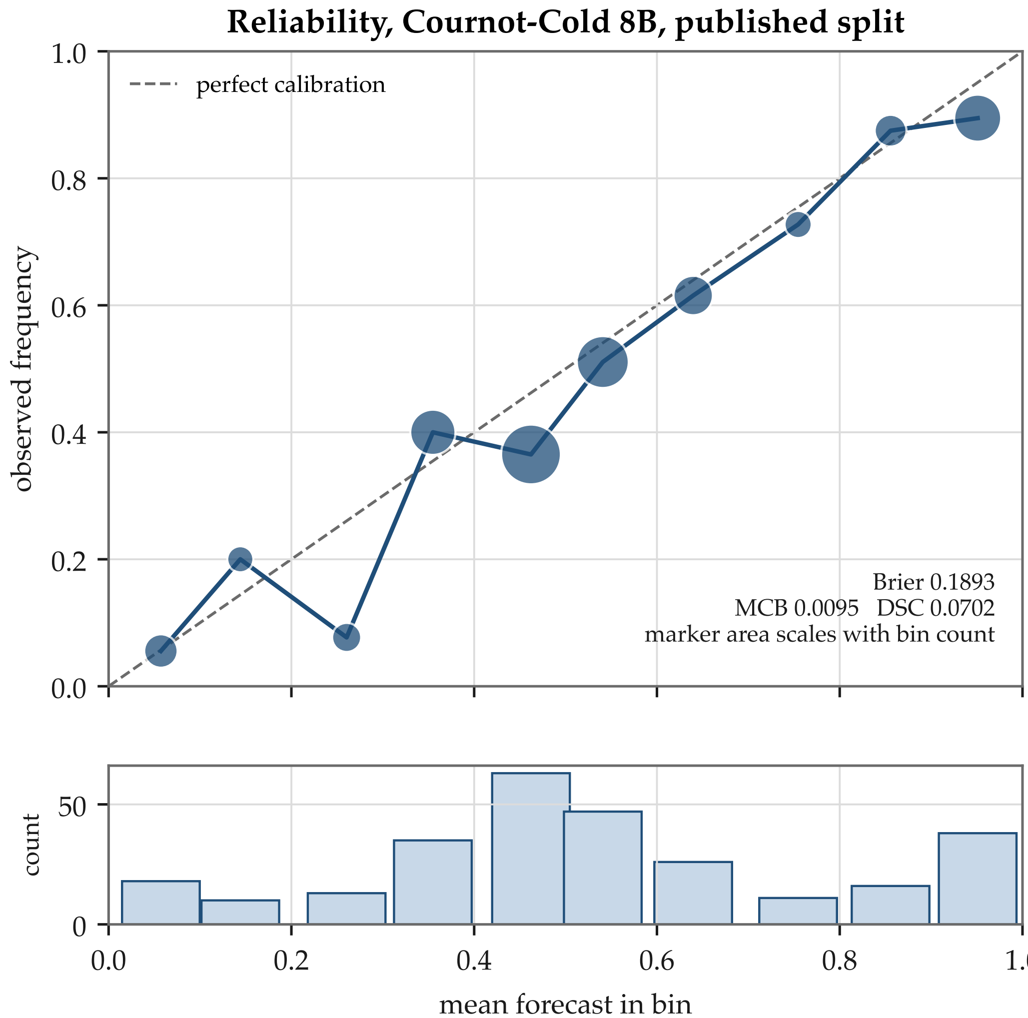

Figure 4: **Reliability for Cournot-Cold 8B on the published split.** Marker area scales with bin count, and empty bins stay empty rather than being interpolated across, so a wobble in a three-question bin cannot read as miscalibration.

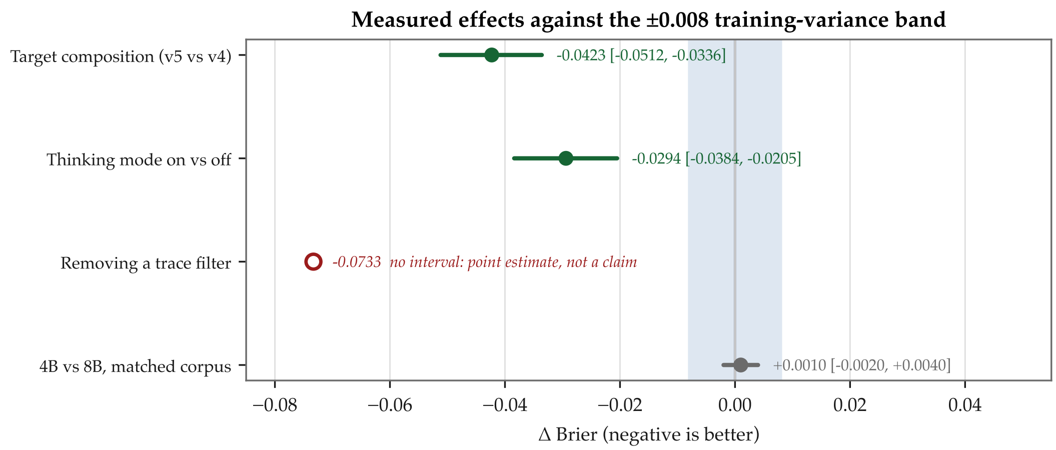

Figure 5: **Every measured effect against the training-variance band.** The shaded region is the interval inside which a difference between two separately trained models is unresolved. The open marker carries no interval and is therefore a point estimate rather than a claim.

#	change	Δ Brier [95%]	Δ resolution [95%]	judged against	verdict
1	**Removing a trace filter** (7,381 filtered → unfiltered)	−0.0733 †	+0.0472 †	question-sampling	large, and the filter had been keeping the traces the teacher got right
2	**Target composition, 0% versus 50% extreme targets** (v5 vs v4)	−0.0423 [−0.0512, −0.0336]	+0.0042 [+0.0011, +0.0076]	question-sampling	significant on both
3	**Serving-stack thinking mode, on versus off**	−0.0294 [−0.0384, −0.0205]		question-sampling	**larger than any training arm in this project**
4	**Scalar head versus text SFT** (0.2329 → 0.1674 dev, best to best)	≈ −0.066	+0.0718 vs +0.0110	mixed classes	the architectural change; not a clean paired interval
5	**Corpus scale, 20,000 → 81,870 terminal rows**	0.1813 → 0.1674	0.0575 → 0.0718	training-run	≈ −0.014, roughly 1.7× the ±0.008 threshold, so it survives
6	**Terminal 0/1 target versus debiased market price**	−0.0035 after calibration	+0.0030	question-sampling	inverted the project's own guiding prior (§6.2, mechanism D)
7	4B versus 8B at matched corpus	+0.0010 [−0.0020, +0.0040]	−0.0011 [−0.0042, +0.0020]	training-run	**equivalent within δ = 0.008**

† **Row 1 carries no interval**, which by this report's own rule (§3.3) makes it a point estimate rather than a claim. It is listed because it is the largest effect the project measured, and omitting it would be a selection. It is marked so that it cannot be read as an established result. Every other row either carries its interval or is explicitly labelled a cross-class comparison.

Row 3 is at its measured rank and stays there. The single largest measured effect in this project is a serving-configuration flag, namely whether the base model's thinking mode was on, rather than any training decision. Demoting it because it is embarrassing would be exactly the selection this report argues against. It is also why a run manifest records six decode fields with no defaults. The same weights, on the same questions, produced calibrated Briers of 0.2288 and 0.2374 depending only on fields that nothing was recording, and a day was spent attributing that difference to training, then to padding, before the cause turned out to be a default nobody had written down.

Dispersion alongside the mean. A mean Brier hides the fact that a model can be better on a minority of questions and worse overall. Where a comparison matters we report the share of questions on which each arm wins alongside the mean, after ForecastBench's leaderboard [15]. This is the single most under-used reporting idea we found in the exemplar sweep, and it is cheap, because it comes free from the paired bootstrap that is already running.

What did not move. No arm in this project ever produced a significant change in resolution against the thinking base on the text lane. The four measurements are v5 at $+0.0014$, v6 at $+0.0000$, v2 at $+0.0018$ and v4 at -0.0032, all null. Four arms, one negative result, reported as such.

6 The failure record

The longest section in this report, and the reason for it.

A note on form before the content. The convention in this genre is a single terminal section, titled something like "Insights from the Unfruitful" [16] or "Unsuccessful Attempts" [17], running to about a page, placed at the end. Both of the best-known examples are visibly under-filled relative to what those labs must have tried, and a terminal section is easy to under-fill because nothing structural demands anything of it. So failures get a numbered body section here, with a fixed schema, in three parts.

The framing for each entry follows the convention those same reports established, and it is worth naming. State the negative flatly, give a mechanism, scope it to this setup, and decline to generalise. That is what keeps a failure record from reading as either self-flagellation or a hedge.

6.1 The failed-arm register

One row per arm. The field that matters most is the third, which records why the arm was expected to work and was written before the result. That field is what turns a list of failures into evidence about judgement rather than a list of things that did not happen.

A rule of admission, stated because it is easy to violate in the flattering direction. This register holds only measured failures, with an interval, on a stated split. Arms abandoned before producing a number are listed separately in Appendix C and are not counted as null results. Promoting an abandonment to a null result is its own kind of overclaim.

arm	what was tried	why it was expected to work (recorded before the result)	outcome	compute	verdict
v1 / arm0	Text SFT, 55,394 examples, 50% terminal targets	Format-locking on a large corpus should teach the model to reason and then commit to a number	**Collapsed.** 392 of 490 forecasts at exactly 0.0; Brier 0.3390	~$5	A corpus defect and a decoder failure together; the second half went unexamined for a day
v4	Thinking-trace distillation, 7,381 traces, `--soft-fraction 0.50`	Distilled reasoning should transfer the teacher's judgement	**Invalid.** 50.2% of targets at exactly 0 or 1 rebuilt v1's collapse; 2,727 of 3,000 forecasts degenerate; below the untrained base after calibration	~3 GPU-hours	The corpus documentation had forbidden this in capital letters, citing v1. A rule in a document is a rule nobody enforces at the moment it is broken
varied `as_of`	φ sampled in [0, 0.9] through each question's life, corpus size held at 20,000	Training across horizons should teach the model to use `as_of`, or at minimum to be invariant to it	**Failed its pre-registered gate.** Horizon drift did collapse, from +0.0125 to +0.0001, but resolution fell significantly at every φ, between −0.0074 and −0.0169, and the histogram narrowed	~1 GPU-hour	The gate had been specified as flat Brier **with resolution held**, which is the only reason this reads as a failure rather than a success
`mixed_full` ("Cold 8B Wide")	95,211 rows across both venues, as a release candidate	Cross-venue training should buy robustness at little cost, since the 13k+13k pilot said it was free	**Declined.** Manifold +0.0056 [+0.0024, +0.0087] worse, significant. Polymarket −0.0223 better on an evaluation set with no builder, and −0.0144 [−0.0225, −0.0064] on the rebuilt one. Decomposed, that is calibration −0.0115 [−0.0224, −0.0014] significant and resolution +0.0067 [−0.0019, +0.0149] null	~$2	The thing it does better is the thing a post-hoc map does. Its Brier lower bound of 0.0064 also sits below the 0.008 training-variance floor. The pilot result was a floor effect
`pmonly`	13,341 Polymarket rows only	A venue-matched model should transfer	Manifold Brier 0.2781, BSS −**16.2%**, sd 0.081, 47 distinct values	~$0.30	Collapsed toward the base rate off-venue
batch7 arm 2	Intended as a seed replication of the filtered terminal arm	A second seed should reproduce within ±0.003	**Not a seed replication at all.** Trained on 7,500 unfiltered rows against a 7,381 filtered comparator; interval −0.0523 [−0.0632, −0.0416]	~$0.30	Caught because the effect was 17 times the recorded seed floor and implausible on its face. A project without a measured floor would have published this
run 4 (4B, 81,870)	The decisive arm of its batch		**Killed at 2h01m having produced nothing**, roughly four minutes from finishing	~$0.53	A sequencing error: the optional arm was chained ahead of the decisive one. This produced the `ops/tooling` directly

Two arms are equivalence results rather than failures, and they are kept out of this table so the distinction stays visible: the 4B, equivalent to the 8B within δ, and the cross-seed replications of both shipped models, both null.

6.2 Corrections, grouped by mechanism

Fifty-one corrections and retractions are recorded, dated, in the research repository's decisions log, across seventeen days and 320 commits. The useful claim is not that we made fifty-one mistakes. It is that they came from six places, and four of the six are addressable by construction rather than by attention.

Each mechanism below carries a **detection field**, which records what caught the failure or admits that nothing did. That field is what makes the measurement-discipline claim demonstrable instead of asserted. **Figure 6** plots the distribution.

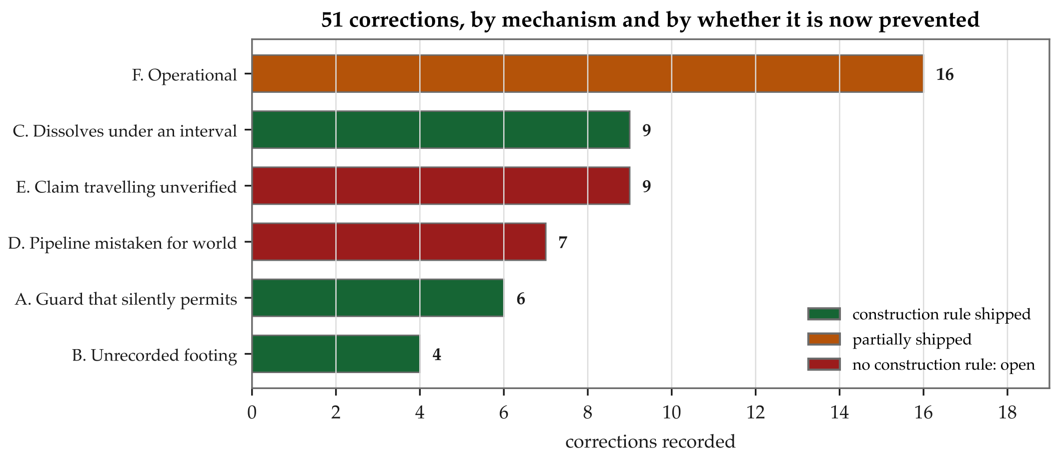

Figure 6: **Fifty-one corrections, by mechanism.** Colour records whether a construction rule now prevents the mechanism or whether it remains open. Two of six are open, accounting for 16 of the 51.

Mechanism A. The guard that silently permits. *Six instances.*

A branch that produces a plausible value where it should refuse. This is the project's signature defect.

- $\text{outcome} = 1 \text{ if winner} == \text{"yes" else } 0$. Polymarket markets are not all Yes/No; 57% of a sampled thirty carried Over/Under, Up/Down, or team names, for which the winner is never "yes". **2,716 of 4,000 rows were silently labelled outcome 0.**
- A reproduction gate whose else-branch printed SKIP and continued when its reference file was missing. The release would have shipped with the gate never having executed.
- A market-metadata API that ignores its filter parameter and returns HTTP 200 with twenty arbitrary markets. A bulk fetch would have joined 160,000 rows to

the wrong questions with no error anywhere.

- An "N/A" string sentinel evaluating truthy, which made a feature constant-True. The null result was the bug. This was the second time that sentinel caused a failure.
- A parity script that silently dropped 339 of 629 rows, computing a table on 46.1% of the set with the count going only to stderr.
- Binning that could be silently ignored while the identity still closed.

> ***Detection.*** *In four of six, nothing detected it, and it was found by writing the thing that would have depended on it. In one case the detection was a push rejection from a file-size limit.* ***Construction counterpart, shipped.*** *Required arguments with no defaults, such as* --min-coverage*,* --terminal-fallback *and* --base-model-release*. Refusal instead of a default in the fetcher. And a* total() *method that raises* TypeError*, because the question it would answer is the one that produced two of these six.*

Mechanism B. The footing that was not recorded. *Four instances.*

Two runs that differ in how they were served are not comparable, and for weeks nothing in the pipeline could express the difference.

The original instance: training ran on raw prompt text while the base model was scored through a server that applies a chat template. Every base-versus-adapter table in the project therefore compared a templated base against raw-prompted adapters. It survived weeks because each script was individually reasonable and no artifact recorded prompt construction. It cost a headline conclusion, since "SFT did not move resolution" turned out to be an artifact of the mismatch.

The second instance, the same day: thinking mode is on by default in one serving stack and off in another. Same weights, same questions, calibrated Brier 0.2288 against 0.2374.

> ***Detection.*** *The first was found by re-scoring on a stack that has no padding axis, while chasing a different hypothesis entirely.* ***Construction counterpart, shipped.*** *A* DecodeConfig *with six fields and no defaults, required on every run manifest, with an executable pre-flight gate that exits non-zero on mismatch. Every value that has ever caused a silent divergence here was a default.* ***Residual, stated.*** *The type could not describe a scalar head until 2026-08-27, which means every Phase 2 run's manifest was unbuildable and the gate refused those runs as malformed rather than checking them. The footing was not mis-recorded. It was unrecordable.*

Mechanism C. The claim that dissolves under an interval. *Nine instances.*

The project's characteristic failure, and the reason the interval gate exists.

- "Qwen3-4B beats Qwen3-8B" became -0.0189 [-0.0444, $+0.0061$], spanning zero.
- "SFT improved resolution by 86%" became $+0.0070$ [-0.0047, $+0.0192$], spanning zero.
- "The published split's date leak inflated resolution" became $+0.0070$ [-0.0155, $+0.0330$], null and underpowered.
- "The 50/50 ensemble beats both components at every horizon, 0.1497 $\rightarrow$ 0.1041" was false in both halves. Those two figures are the first and last cells of a single column, presented as a before and after. And the ensemble is null at $\varphi = 0.75$ and significantly worse at $\varphi = 0.90$, at $+0.0068$ [$+0.0017$, $+0.0119$].
- "The model matches the crowd at half-life" became a crossover at $\varphi^* = 0.390$ [0.27, 0.51], with the model already behind by $+0.0109$ at $\varphi = 0.50$.

> ***Detection.*** *The gate, in six of nine. In two of those six the story had already been told before the interval was run. In the published-leak case the mechanism was narrated to the project lead one message before the number existed. The gate caught it. Telling the story first did not help.* ***Construction counterpart, shipped.*** *A procedural gate that forbids the words "beats", "improves", "better", "worse" or a percentage change without an interval, and that requires a power check before any null is believed.* ***Residual, and it is the largest one in this report.*** *The interval that gate computes is the wrong variance for most of what it was used on. See §3.2. The gate is correct, and it was being applied to comparisons it does not cover.*

Mechanism D. Reasoning from what we measured to what is true. *Seven instances.*

Inferring a property of the world from a property of our own pipeline.

- "At $\mathrm{as\_of} = \mathrm{open\_date}$, 1 of 3,000 questions has a market price at all" was an artifact of reading at exactly t = 0. By $\varphi = 0.10$, 96% have traded.
- "The corpus has price history for Manifold only" was false. The other venue's price history is 5.1 GB on disk, and one evaluation file carries a price on 15,000 of 15,000 rows. This one did not produce a wrong number. It produced a year of not running an experiment we could already run.
- Four arguments against pursuing an evidence channel, of which three were about our own pipeline rather than about evidence.
- "The model uses $\mathrm{as\_of}$ backwards" became "the model moves when it should not." Zero of 3,000 evaluation questions resolve early, so with no evidence channel the Bayes-optimal response to a later $\mathrm{as\_of}$ is no movement at all. That reframing is what turned the varied-$\mathrm{as\_of}$ gate into one that could fail.

- The project's own guiding prior, which was to prefer the low-variance target, inverted on this task. The maximum-variance terminal 0/1 target beat the low-variance market-price target on resolution, because the price target is biased. The prior survives only in its qualified form: prefer the lower-variance target among unbiased ones.

> ***Detection.*** *Adversarial review by a second reader, in five of seven.* ***Construction counterpart: none.*** *This is the mechanism least addressable by construction, and the honest statement is that we have none for it. The nearest thing is a standing rule created after one instance, requiring that an evidence-utilisation arm include a destroyed-information control in the same run. That rule exists because a reviewer asked for a shuffled-values control in a single line, and the control returned −0.0032 [−0.0552, +0.0489]. The model responded identically to a scrambled series, and two conclusions were withdrawn. Rigour had been applied to the implementation and not to the design.*

Mechanism E. A claim travelling without verification. *Nine instances.*

Repeating a stated fact instead of checking it. This group includes the only fabrication in the record.

A citation attributed a specific diagnosis and two specific figures, 17.09% and 57%, to a named paper. The attribution was wrong, the paper does not contain the string in question, and neither figure appears in either candidate paper. The real figures, from the paper that was actually meant [18], are more damning than the invented ones: 71% and 81% major post-cutoff leakage, 41% and 55% answer directly revealed, and a Brier of 0.242 leak-free degrading to between 0.108 and 0.128 when leaky.

That is the uncomfortable part, and it is the generalisable lesson:

> ***The fabrication was not load-bearing. It supported a correct conclusion. That is exactly why it survived, because a wrong number that supports a conclusion you already believe gets no scrutiny.***

A second instance the same day: a second fabricated figure was sitting beside the first one's retraction marker and was not caught by it, including by the person who had written that marker hours earlier. A retraction marker is a signal to audit the surrounding claims, not only the marked one.

A third instance is structurally identical and happened inside a correction. A claim was corrected in two files on the strength of a handed list, without sweeping. It was in four. The correction reproduced the failure it was correcting, about half an hour after the common factor had been written down.

A novelty sweep in the same period killed four of five stated novelty claims, including one that had been recorded as a product differentiator. One survived.

> ***Detection.*** *An external adversarial pass, in seven of nine.* ***Construction counterpart: owed and not written.*** *Nothing structurally prevents this. The nearest*

available rule is that a retraction marker triggers an audit of its neighbourhood, which is a detection rule, and it has now fired twice.

Mechanism F. Operational, where nothing was measuring. *Sixteen instances.*

The largest group, and the one with the clearest self-diagnosis.

- **A pod ran for 15.0 hours to do 6.2 hours of work, and roughly $3.90 bought nothing.** Three independent safety layers had been designed. The on-pod deadline did not stop the meter. The external watchdog failed on a date format, because the vendor returns Go `time.Time` formatting rather than ISO 8601, and since it had been written to keep a pod on unparseable input, it logged the identical line thirty times over nine hours. What actually caught it was the project lead asking whether the job was done.
- **The 1.7B's weights were destroyed before a reserved ship-or-no-ship decision had been made.** A recommendation was recorded and the artifact it concerned was deleted in the same action. It was recoverable only because the run could be repeated, which is how we discovered that it could not be. See §3.2.
- **A completed variance run was lost** thirteen hours after the watcher printed "FINISHED, download and TEAR DOWN". A pod was terminated from a web console. The guard that refuses teardown until collection succeeds worked exactly as designed. It was simply not on the path taken.
- **Buffering, twice, in two different layers.** Consumer-side, through `| tail`, and producer-side, through Python's own stdout buffer. The second recurred six hours after being written up.
- `pgrep -f` **matching the watcher's own command line**, so a wait loop waits forever for itself.

Detection. *A human asking a question, in the two most expensive cases.* ***Construction counterparts, shipped.*** *Teardown now refuses until collection exits 0 and verifies byte size against the remote. The collector enumerates what is on the pod instead of printing a suggestion. A* `.gitignore` *pattern was generalised after a third near-miss. An undatable pod now alerts and exits non-zero rather than being silently kept.* ***The self-diagnosis is the part worth quoting.*** *The lesson is not a fourth layer. It is that the rule "a guard never observed to fire is not known to work" was applied religiously to metric code and never to operational tooling. Three guards fired for real in one week, all of them written for branches that had never executed, and one of them caught something larger than the release it was guarding.* ***Two rules were earned here in two days, in two different scripts.*** *A printed suggestion is not a safeguard: the first time it cost the weights, the second time it cost a run. And a guard on your own script does not cover the console.*

6.3 What the distribution says

mechanism	count	detected by	construction counterpart
A. Guard that silently permits	6	mostly nothing; found downstream	**shipped**
B. Unrecorded footing	4	chasing a different hypothesis	**shipped**, with a stated residual
C. Dissolves under an interval	9	the gate, 6 of 9, twice too late	**shipped**, and itself corrected in §3.2
D. Our pipeline mistaken for the world	7	adversarial review, 5 of 7	**none; open**
E. Claim travelling unverified	9	external audit, 7 of 9	**owed, not written**
F. Operational	16	a human asking, in the two costliest	**shipped, partially**
total	**51**		**4 of 6 addressed**

Three things follow from this distribution that do not follow from the individual entries.

First, the denominator is the argument. A survey of 316 scientists [19] found that roughly 12% had lost confidence in one of their own published papers for reasons they took personal responsibility for, that the loss was made public in 17% of cases, and that it reached a formal retraction or correction in 3 of 24. A project reporting 51 dated corrections is not confessing. It is reporting a rate that almost nobody reports at all. The relevant published finding is that willingness to admit error is not a reliable indicator of propensity to commit error, since errors are frequent throughout the record, and that public admission is better read as a credible signal that the issuer values correctness.

Second, the biggest group is the one nobody was measuring. Mechanism F is 16 of 51, roughly a third, and it is entirely operational. The project applied a rigorous testing standard to metric code: 1,117 tests across 40 files, a maintained register of 43 named mutations each paired with the test that killed it, and a test suite of 10,188 lines against a library of 8,459. It applied almost none of that to the tooling that spends money and holds artifacts. The `ops/` directory has no tests. That asymmetry is the clearest actionable finding in this report, and it is still open.

Third, and least comfortably:

> ***A discipline of recording failures is not the same as a discipline of preventing them, and the first is much easier to sustain.***

Mechanisms D and E account for 16 of 51 between them and have no construction counterpart. They were caught by a second reader, repeatedly, and the project has no mechanism that would have caught them without one. Every entry in those two groups is evidence that the recording discipline worked and the prevention discipline did not exist.

6.4 The pre-registered kill

Three things were registered before their experiments ran. All three fired. Two fired against the hypothesis. None was re-specified afterwards.

Registration 1, the varied-as_of **arm.** The registered gate was flat Brier across the φ sweep with resolution held, and explicitly not lower Brier. It was registered that way because the failure mode had been named in advance: the arm could improve aggregate Brier purely by hedging late-life forecasts toward the base rate, and that is not skill. The arm failed. Brier drift collapsed as predicted, from +0.0125 to +0.0001, and resolution fell significantly at every φ. Had the gate been lower Brier, this arm would have shipped.

Registration 2, the contamination inference. A resolution threshold of 0.05 was registered before the transfer numbers were seen. The measurement came in at 0.0165, below the line. The registered escalation was nonetheless declined, because reading the question text afterwards showed that the registration itself was flawed: it treated venue as the only variable when question composition differed too. Both the failure and the decline are recorded. The registration still did its job, in that it stopped the number being read as whatever was convenient afterwards.

Registration 3, the reproduction gate. Tolerance 0.02, written before any data existed, on a hunch, for a branch that had never executed. It fired, at a maximum per-question difference of 0.6793, and what it caught, described in §3.2, was larger than the release it was guarding.

The leak this experiment nearly had. A separate pre-registered experiment on entity reference classes measured its own leakage surface before running. A tracer case showed 17.6-fold pageview inflation after the event it was supposed to predict. The naive version of that experiment, which is the version that would have been run had pre-registration not forced the question, would have measured that inflation and reported it as signal.

One honest note on registration counts. These three are the registrations that were written down. The project did not maintain a registry of all planned analyses, so the deviation rate, meaning the fraction of analyses that departed from a pre-specified plan, is not knowable here. Published audits put the rate of unregistered analytic steps in pre-registered studies at around 90% [20]. **UNVERIFIED**: that figure is quoted from a secondary summary, because the primary source is paywalled and was not read. We have no grounds to claim we did better. We have grounds to claim that three specific gates were fixed in advance and honoured.

7 Cost, honestly

The convention in this genre is to report the main training run's compute and exclude prior research and ablation experiments [21]. We reject that convention explicitly, following the accounting in [22],, because at this scale the excluded number is the interesting one. A \$62 total is a claim about measurement efficiency, and it is only credible if the failed runs are inside it.

Hardware. A single A40 with 44 GiB at $0.44 per hour for every batch run. An A100 SXM 80GB at $1.39 per hour was priced and declined. No network volume was used, deliberately, since a LoRA adapter is between 150 and 350 MB and goes to container disk. Two 30 GB volumes were found billing with nothing attached, which is how that decision came to be made.

8B training fits in roughly 30 GB at batch size 4 and runs out of memory at batch size 8. It does so at step 800 of 922, which is late enough to waste the entire run, because nothing is saved until the end.

category	cost	share
Shipped models' training runs (8B 3h31m, 4B 2h34m, 1.7B 1h32m twice)	~$4.20	7%
Ablations and arms that did not ship	~$14	23%
Scoring, transfer, published re-scores	~$1.50	2%
Runs that produced nothing usable	**~$0.83**	1%
Idle pods, billed after the work finished	**~$3.90**	6%
Remainder (bake-offs, probes, batches 6 to 11, rolling)	~$31	51%
API spend (baselines, teacher pilots, adversarial material)	~$6.20	10%
Total	**~$62**	

How much of that table to trust. The total and four of the rows are recorded figures: runs that produced nothing at ~$0.83, idle-pod time at ~$3.90, API spend at ~$6.20, and the shipped runs' wall-clock at the quoted A40 rate. The split between shipped models and ablations is reconstructed from per-batch costs and carries some double-counting, because one shipped arm was trained inside a window whose cost was recorded only in aggregate. The remainder row is a residual rather than a measurement. It is reported this way rather than smoothed, because a clean-looking decomposition would be the less honest artifact. **Figure 7** shows the breakdown with the two reconstructed rows marked.

The two rows that matter are the two that bought nothing. Roughly $0.83 for a run killed about four minutes from completion, and roughly $3.90 for a pod that ran nine hours past its work. Together they are 7.6% of the budget, and the second is larger than the entire cost of training all three shipped models. That ratio is why §6.2 mechanism F accounts for a third of the correction record.

Wall-clock, including crash resumptions. Seventeen days end to end, and 320 commits. The longest single batch consumed 15.0 hours of billing for 6.2 hours of work. One arm died at 2h01m with nothing to show. A generation run wedged at 16,083 of 20,000 prompts for 33 minutes before a stall deadline existed, the root cause being that single-GPU serving has no internal stall timeout and the two environment variables that appear to provide one are declared and never read. The lesson recorded at the time: a progress bar is a completion counter, never a liveness signal.

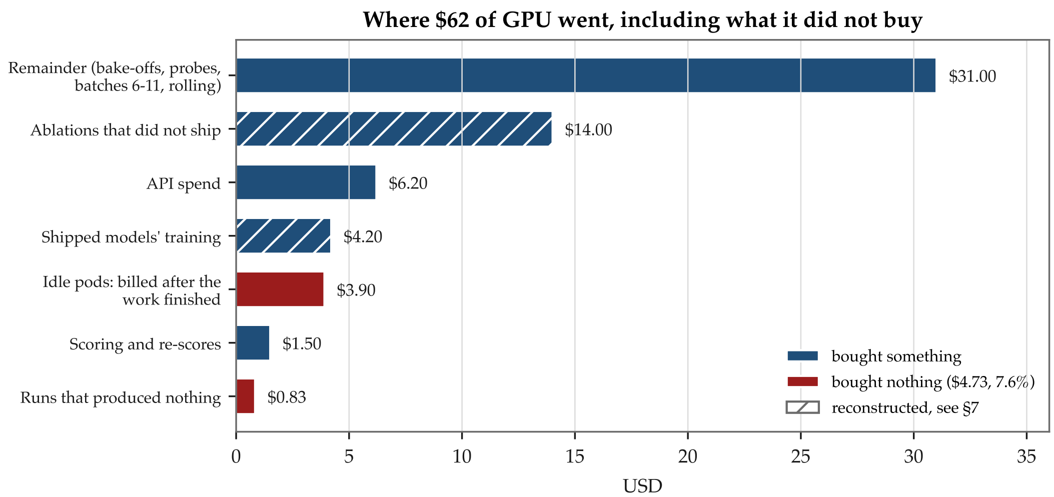

Figure 7: **Where $62 of GPU went, including what it did not buy.** The two red rows bought nothing, and together they exceed the cost of training all three shipped models. Hatched rows are reconstructed rather than recorded.

Cost per forecast, for scale. Qwen3-8B at bf16 costs roughly $0.00004 per forecast against a frontier API baseline at $0.017, a ratio of about 500. That ratio, rather than the Brier, is the case for a small calibrated model.

Runs that produced no usable result: three. Run 4, killed at 2h01m. Batch7 arm 2, which trained on the wrong corpus. Batch11 run 4, which completed and was never collected.

8 The forward track record

Every published forecast in §4 was generated after the question had resolved. The as_of was set correctly and the split is contamination-free by construction, so the numbers are sound. But a reader is being asked to accept that we set as_of honestly.

A forecast recorded before the outcome exists asks them to accept nothing.

Questions captured while open	**6,734**
Forecasts recorded before any outcome existed	**6,734**
Resolved so far	**3**
Coverage	**0.0004**
Excluded, resolved between capture and forecast	4
Still open	6,726

Zero coverage is the expected and correct state one day into a forward record whose questions were frozen with horizons of at least 48 hours. The value accrues over weeks and cannot be backfilled. Publishing it now, at a coverage of 0.0004, is the point, because the only wrong move is not starting.

How it is made checkable.

Platform-owned timestamps. The question's creation and close times come from the venue rather than from us. Our own capture time is recorded separately so the two cannot be confused.

A new dated file per capture, never rewritten, plus a content hash over the question ids. Git history, a dated filename and a content hash are three independent ways to catch a retroactive edit, which is two more than a claim.

Payload in object storage, hash in git. Neither alone is enough. An object can be overwritten and the replacement looks identical, while 2.8 MB a day is about a gigabyte a year and git will not carry it comfortably. A hash committed in 2026 against an object fetched in 2027 either matches or does not, and the hash was published before the outcome existed.

Three ordering checks, each a refusal. Resolution must be strictly after capture, and violating that refuses the whole run, because it means the platform contradicts its own resolved flag. The forecast file must predate every outcome it is joined to, and violating that excludes the row and continues, because it is a property of our scheduling rather than of the data. And a capture is never rewritten.

Unresolved questions are retained with `outcome: null` **rather than dropped.** Dropping them would let the record quietly become "the questions that resolved quickly", which correlates with horizon, which is the axis this model's skill varies along most. The resulting bias would be invisible in the final Brier.

Two limits, stated plainly.

This is detection, not prevention. It is not object-lock and does not claim to be. It makes a retroactive edit loud rather than impossible.

There is a structural gap between capture and forecast, and it is not an oversight. Capture is CPU-only and runs unattended weekly. Forecasting needs a GPU, which costs money, and a standing guardrail requires per-job authorisation for anything that spends. An unattended job that provisions a pod is exactly what that guardrail forbids. So the gap is structural, and the honest response is not to close it but to make it visible while it is still cheap to act on. It has already cost one question: a capture ran at 01:07 and the forecast at 22:05, and one question resolved in the 21-hour gap. The ordering check caught it, correctly. A second layer re-checks freshness before the GPU spends anything, but it checks the scheduled close time, and questions can resolve early, which no lag computed from a close time can anticipate. Only the after-the-fact check is a correctness guarantee.

9 What we do not claim

Numbered, in the body, as its own section. This is the section the genre deletes as reports get longer. Two of the largest open-model reports in the current literature, at 92 [23] and 118 [24] pages, have no general limitations section at all. Limitations survive peer review and disappear from self-published reports.

1. **Not state of the art.** See §0 and §1.
2. **No claim of a 4B win.** The 4B is equivalent to the 8B within $\delta = 0.008$ on dev, and the published-split comparison is underpowered rather than equivalent. See §3.5. Those are different findings and are not interchangeable.
3. **Calibration does not transfer across venues.** ECE is 0.023 on the training venue's dev split against 0.156 on a different venue. A new venue needs its own calibration mapping, and we have not built one.
4. **Transfer is not uniform, and the aggregate hides it.** On the cross-venue transfer set the 8B scores a Brier of 0.1780 and a BSS of +2.2% overall. Politics, at $n = 209$, scores 0.0789 with a BSS of +56.0%, while Elections, at $n = 461$, scores 0.2278 with a BSS of −**23.1%**, which is materially worse than a constant. Those questions name obscure local candidates, which is a lookup rather than a forecast. The aggregate transfer number should never be quoted without both halves. **This number also carries the defect described in §9.1 and should not be used at all until that is resolved.**
5. **A single** as_of **per question.** The shipped evaluation forecasts at each question's open date. The one arm that trained across horizons failed its gate, as described in §6.1. We do not know that the model uses as_of at all in any useful way.
6. **A single model, with no ensemble.** The one ensemble claim the project made was retracted in full, in §6.2 mechanism C, and the ensemble is significantly worse at the longest horizon tested.
7. **The training-variance mechanism is unverified.** Non-deterministic CUDA kernels is a hypothesis. **UNVERIFIED.**
8. **The bootstrap may undercover at $n = 277$.** See §3.3. **UNVERIFIED.**
9. **The correction record is a lower bound.** §6 lists what was caught. The direction of the bias is knowable, since 16 of 51 were caught only by a second reader, and there is no reason to believe a second reader caught all of them.
10. **The forward record currently establishes nothing about accuracy.** There are three resolutions. It establishes only that the forecasts existed first.
11. **This report has not been externally reviewed.** Every audit described in §6 was internal or performed by an adversarial agent, not by peer review. The genre's own evidence, cited in the opening of this section, is that limitations sections survive peer review and are deleted without it.

9.1 One open defect in the shipped artifact, stated in full

This report would be more flattering without this subsection. It is here because the defect is live, it sits underneath numbers that are already public, and a disclosure a reader has to reconstruct from a footnote is not a disclosure.

What is wrong. The Kalshi transfer evaluation set, kalshi_score_778.json, carries a filter field naming a selection rule. **No committed code in this repository implements that rule.** The file's metadata records no sampling procedure, no seed, and no reproducible build path. It is the third instance of a pattern this project has now hit three times, in which an evaluation artifact exists with no builder behind it.

Why it matters more than the other two. The first instance, the published split, was caught and rebuilt, and rebuilding it revealed a genuine leak: the scheduled resolution date equalled the actual resolution date on 132 of 132 published questions against 37.7% on dev, so every published question had been telling the model the date it actually resolved. The second instance, a Polymarket scoring set, was caught before any release depended on it, and the arm that would have depended on it was removed rather than repaired.

This third instance is different, because it already shipped. The Kalshi transfer numbers sit under a transfer section on all three public model cards. The numbers themselves are real forecasts from the correct models, recomputable by anyone from the shipped files, and verify.py reproduces them exactly. What cannot be reproduced is the question set they were computed on.

What that does and does not invalidate. It does not make the forecasts wrong. It does not affect any number on the published split, which is the only source of a headline figure, and which has a builder, a content hash, and a full drop accounting. What it means is that the transfer claim rests on a population nobody can rebuild, so an independent party cannot check whether that population was selected in a way that flatters the result. That is not an accusation against the number. It is the reason the number is not yet evidence.

The distinction is worth stating precisely, because it is the same one this project has drawn before: **the set is not wrong, it is unreproducible, and for a number bound to a model card those amount to the same problem wearing different clothes.**

What we are doing about it. A registered builder for the Kalshi set is the next item of work, and the rule it implements will be written before it is run and will not be adjusted afterwards, following the procedure used for the Polymarket rebuild. Until that exists, the transfer numbers in §9 item 4 and in verify.py output should be read as descriptive rather than as claims, and this report does not rest an argument on them.

Why we are not simply removing them. Deleting the transfer section would make the cards look better and the record worse. The unflattering half of the transfer result, the Elections BSS of -23.1%, is the part a reader most needs, and removing the section to avoid disclosing a defect in its provenance would be a selection of exactly the kind §5 and §6.1 argue against.

10 Reproducing every number

verify.py recomputes every headline number in every model card from the shipped forecasts, with no model, no GPU and no network.

```
uv run python verify.py
```

Verified output for the 8B, run offline on CPU during the compilation of this report:

```
PUBLISHED (headline, contamination-free)  n=277
  Brier          0.1893
  base rate      0.4946   base-rate Brier 0.2500
  calibration    0.0048   (lower is better)
  resolution     0.0627   (higher is better)
  ECE eq-width/10  0.0558   eq-mass/10 0.0525
  CORP  MCB 0.0095  DSC 0.0702  UNC 0.2500   (bin-free; residual 0.0e+00)
  distinct values  83   sd 0.2502
  BSS vs base rate +24.3%  [diagnostic, not comparable across corpora]
```

Those figures match the model card's metadata block exactly, and the same holds for all three models. If a number in verify.py's output disagrees with the model card, the model card is wrong.

verify.py also prints the unflattering numbers rather than omitting them. The Elections BSS of −23.1% appears in its standard output with its interpretation inline, subject to the provenance caveat in §9.1.

Independently reproduced during this report's compilation:

claim	method	result
1,117 tests across 40 files, all passing	ran the suite	confirmed
harness is 32 files, 8,459 lines	counted	confirmed
exactly one third-party dependency	grepped every top-level import	confirmed, pydantic only
card headline numbers	verify.py against card metadata	**exact match, all three models**
CORP residual is zero	verify.py	0.0e+00 on every split
published split content hash	recomputed from the shipped file alone	**match**
the card certifies	ran certify_card.py	**CERTIFIED**
training corpus row count	wc -l	81,870

What a third party can check with no weights, no GPU and no network: every headline number on all three cards; the transfer numbers including the bad ones, subject to §9.1; that the evaluation set's content hash matches its own contents; that the split is post-freeze by construction, since the freeze and every resolution time are in the file; the full drop accounting; and the complete harness source, so the metric definitions are auditable rather than asserted.

What they cannot check, stated because it is the honest boundary:

1. **That the forecasts came from the claimed weights.** verify.py reads a JSONL. The binding of forecasts to an adapter rests on the run manifest and the Hub upload. Re-deriving them needs the adapter, a GPU and the base model.
2. **That the harness is correct.** The 1,117-test suite is not shipped in the release repositories. It lives in the research repository.
3. **The question text**, deliberately, per §1. It is retrievable from each venue by id.
4. **Reproducing training.** The recipe is pinned. The outcome is not, per §3.2.
5. **The Kalshi transfer population**, per §9.1.

Prompts, and how much they were engineered. The exact prompt construction ships in scalar_score.py. Prompt engineering against the metric was not performed on the shipped models: the corpus builder passes question, criteria, scheduled date and as_of, and that string has not been tuned against dev Brier. One prompt was tuned in an unrelated exploratory arm, and there it moved MCB from 0.0641 to 0.0136, which is a useful indication of how large this unmeasured axis can be.

One caveat on the "no dependencies" claim. verify.py itself uses only the standard library, but importing the harness package pulls in pydantic. The documented uv run invocation installs it. A bare system interpreter fails.

11 Appendices

A. The decisions log's public face

The log is not published in full. Two load-bearing properties of it are published here, because they are what make §6 evidence rather than narration.

Retractions stay in place. Superseded entries are struck rather than deleted, and the superseding entry is dated. The 2026-08-13 freeze entry, which was backdated by about 28 hours and superseded within a day, is still there.

Pre-registered entries were written before outcomes were known, and their ordering in the log is checkable against commit timestamps.

At the time of this report the log runs to 14,063 lines and ends at entry 2026-08-29a. A stated count of 135 entries appears in the repository README, and the two have not been reconciled. **UNVERIFIED.**

B. Run manifests and DecodeConfig

Every evaluation run writes a manifest with seven mandatory fields: model hash, data snapshot hash, evaluation split, git SHA, leakage-detector version, freeze, and decode configuration. The git SHA refuses a dirty worktree, because a SHA recorded next to uncommitted changes names code that was never what ran, which is worse than no SHA at all, since it looks like provenance.

DecodeConfig carries the six fields that have actually diverged in this project.

field	why it is here
serving_stack	engines batch, pad and position differently
prompt_construction	chat_template against raw; the original defect, and weeks of invalid tables
temperature	one stack defaults to 0.8 and does not say so
thinking	worth -0.0294 [-0.0384, -0.0205], larger than any training arm here
probability_read	the sign of one arm's effect flips with this, and both signs are significant
max_new_tokens	must be exactly 0 for a scalar head, since a nonzero value would record a footing that does not exist

The concrete demonstration is more convincing than the rule: a manifest that records the model hash but not these six cannot distinguish 0.2288 from 0.2374 on identical weights.

The shipped footing is transformers==4.57.1, chat template, temperature 0.0, thinking off, scalar-head read, max_new_tokens=0, base model released 2025-04-29, freeze 2026-08-15.

C. Arms attempted and abandoned

Listed separately from §6.1 because they produced no number and are therefore not null results.

A seven-arm λ sweep over supervision-target mixtures, fully specified with volume-equalised arms and a Holm-Bonferroni correction across seven. Declared unrunnable the same day for want of a base model, training code, a forecaster and compute. Superseded by the Phase 2 re-scope.

A four-rung synthetic-to-real mixture ladder. The library and its event-partition gate were both built. No arm ever ran, because the synthetic corpus's justification collapsed under a shuffled-values control, described in §6.2 mechanism D, before any rung was trained.

RL via GRPO. Deferred and never implemented. The reward-variance thesis that motivated it survived a literature sweep. The delivery mechanism did not.

D. Per-split noise floor

surface	n	quantity	measured floor
dev (Manifold)	3,000	single-run sd, identical recipe, n = 3 runs	**0.0030** (range 0.0053)
dev (Manifold)	3,000	two separately trained models, 95% threshold	≈**0.008**
dev (Manifold)	3,000	paired bootstrap half-width, Brier	≈0.0030
dev (Manifold)	3,000	paired bootstrap half-width, resolution	≈0.004
dev (Manifold)	500	paired bootstrap half-width, resolution	≈0.012, **unmeasurable**
published	277	paired bootstrap half-width, Brier	≈0.011

Every delta in §5 should be read against the appropriate row. The 500-question row is retained because it is the reason the evaluation set holds 3,000.

E. Datasheet pointer

A datasheet ships with each model, following the seven-section convention [25], with the timeframe question answered explicitly because it is the direct ancestor of this project's as_of rule. It records the corpus source, the normalization accounting in which kept plus rejected must equal raw and is asserted rather than assumed, the split boundaries, and the four timestamp invariants each source was checked against.

One of those invariants disqualified two candidate venues: a resolution timestamp must precede the moment the snapshot containing it was taken. One venue failed it on 39% of its resolved rows, 10,619,937 of 27,017,646. That figure is **UNVERIFIED** here, since it requires the corpus dump and was not re-run for this report.

References

Numbered by first appearance. **Every arXiv entry below was checked against the arXiv metadata API rather than written from memory**, because §6.2 mechanism E of this report is a fabricated citation that survived precisely because nobody opened the paper. That check was not ceremonial: it caught two invented titles in the first version of this very section, and corrected a first author, a title ending, and a preprint-versus-journal title mismatch. Entries without an arXiv identifier were taken from the project's literature sweep; where a source was not read in full, the entry says so.

1. N. Chandak et al. *Scaling Open-Ended Reasoning to Predict the Future.* arXiv: 2512.25070. The OpenForecaster 8B paper.
2. M. White et al. *The Model Openness Framework: Promoting Completeness and Openness for Reproducibility, Transparency, and Usability in Artificial Intelligence.* arXiv:2403.13784.
3. E. Miller. *Adding Error Bars to Evals: A Statistical Approach to Language Model*

Evaluations. arXiv:2411.00640.

4. L. Madaan et al. *Quantifying Variance in Evaluation Benchmarks*. arXiv:2406.10229.
5. S. Biderman et al. *Lessons from the Trenches on Reproducible Evaluation of Language Models*. arXiv:2405.14782.
6. P. Henderson et al. *Deep Reinforcement Learning that Matters*. arXiv:1709.06560.
7. S. Bowyer et al. *Position: Don't Use the CLT in LLM Evals With Fewer Than a Few Hundred Datapoints*. arXiv:2503.01747.
8. D. G. Altman, J. M. Bland. *Absence of evidence is not evidence of absence*. BMJ 1995;311:485. PMC2550545.
9. G. Piaggio et al. *Reporting of Noninferiority and Equivalence Randomized Trials: Extension of the CONSORT 2010 Statement*. JAMA 2012. Link.
10. Quality audit of noninferiority and equivalence trial reporting. Trials 2012. PMC3554513. Source of the 94%-defined, roughly-a-quarter-justified figures.
11. A. H. Murphy. *A New Vector Partition of the Probability Score*. Journal of Applied Meteorology, 1973. The calibration / resolution / uncertainty decomposition. Not an arXiv paper and not re-verified here.
12. T. Dimitriadis, T. Gneiting, A. I. Jordan. *Stable reliability diagrams for probabilistic classifiers*. PNAS 2021. The CORP method. The preprint, arXiv:2008.03033, carries a different title: *Evaluating probabilistic classifiers: Reliability diagrams and score decompositions revisited*.
13. A. Kumar et al. *Verified Uncertainty Calibration*. arXiv:1909.10155. Binned ECE as a lower bound.
14. A. Perez-Lebel et al. *Beyond calibration: estimating the grouping loss of modern neural networks*. ICLR 2023. arXiv:2210.16315. Lemma C.5, cited for the resolution-preserving property of a strictly monotone reparameterisation of the log-odds.
15. E. Karger et al. *ForecastBench: A Dynamic Benchmark of AI Forecasting Capabilities*. arXiv:2409.19839.
16. DeepSeek-AI. *DeepSeek-R1: Incentivizing Reasoning Capability in LLMs via Reinforcement Learning*. arXiv:2501.12948. §4.2, "Unsuccessful Attempts".
17. N. Lambert et al. *Tülu 3: Pushing Frontiers in Open Language Model Post-Training*. arXiv:2411.15124. §8.2, "Insights from the Unfruitful".
18. R. Alur et al. *AIA Forecaster: Technical Report*. arXiv:2511.07678. The paper this report once cited under a fabricated attribution and two invented figures; see §6.2 mechanism E.
19. J. M. Rohrer et al. *Putting the Self in Self-Correction: Findings From the Loss-of-Confidence Project*. Perspectives on Psychological Science 2021. Link · open PDF.

20. K. Willroth, O. Atherton. *Best Laid Plans: A Guide to Reporting Preregistration Deviations.* AMPPS 2024. Link. **Paywalled and not read**; the rate is taken from a secondary summary and is marked UNVERIFIED in §6.4.

21. DeepSeek-AI. *DeepSeek-V3 Technical Report.* arXiv:2412.19437. Cited for the convention this report rejects: its compute figure explicitly excludes prior research and ablation experiments.

22. J. Morrison, C. Na, J. Fernandez, T. Dettmers, E. Strubell, J. Dodge. *Holistically Evaluating the Environmental Impact of Creating Language Models.* arXiv:2503.05804. Source of the finding that model development amounted to roughly 50% of the impact of training.

23. A. Grattafiori et al. *The Llama 3 Herd of Models.* arXiv:2407.21783. 92 pages, no general limitations section.

24. Team Olmo, Allen Institute for AI. *Olmo 3.* arXiv:2512.13961. 118 pages, no general limitations section; also the source of the per-benchmark noise-floor idea in Appendix D.

25. T. Gebru et al. *Datasheets for Datasets.* arXiv:1803.09010.

26. Cursor Research Team. *Composer 2 Technical Report.* arXiv:2603.24477. This report's page layout follows it.

What the check caught. Drafting this list from memory produced two wrong titles. Entry 18 was written as *Reasoning Models Are Not Superforecasters*; it is *AIA Forecaster: Technical Report*. Entry 22 was written as *Addition is All You Need: Estimating the environmental footprint of AI*; it is *Holistically Evaluating the Environmental Impact of Creating Language Models*. Both were plausible, both supported the sentence citing them, and neither would have been noticed by a reader who did not open the link. That is the same shape as §6.2 mechanism E, occurring inside the section written to prevent it, which is worth recording rather than quietly fixing.

Provenance of this report

Assembled on 2026-08-29 from four source documents, all in the research repository.

report-source-experiments.md supplied every training arm, every measured number that reached a conclusion, every retraction, and all compute.

report-source-infrastructure.md supplied the pipeline end to end, the harness module by module, the nine numbered operational failures, and the separation between what was verified by execution and what was only read.

report-craft-frontier.md is a structural study of roughly thirty model technical reports and system cards. It supplied this report's section ordering, its changelog placement, its cost-reporting form and its failure-record shape. The page layout follows the Composer 2 technical report [26].

report-craft-rigour.md collects reporting conventions from clinical trials, metascience and engineering post-mortems. It supplied the equivalence margin in §3.5, the question-flow diagram in §3.1, the per-mechanism correction template with its detection field in §6.2, and the denominator argument in §6.3.

The two source records mark what they verified by execution and what they only read. That distinction is carried into this report as the **UNVERIFIED** mark, which appears six times.

Twelve discrepancies between the repository's documentation and its code were found during compilation and are recorded in the infrastructure source. Four bear on numbers in this report and are disclosed where they are used, in §9 items 7 and 8, §9.1, Appendix A and Appendix E. The remaining eight are documentation drift with no effect on any published figure.